Shijun Cheng

Shijun Cheng | GitHub | Google Scholar | shijun.cheng@kaust.edu.sa

Education

2018.09 – 2022.01 **Ph.D., Solid Geophysics**
University of Chinese Academy of Sciences.

2015.09 – 2018.06 **M.Sc., Applied Geophysics,**
Jilin University.

2011.09 – 2015.06 **B.S., Applied Geophysics,**
Jilin University.

Research Positions

2025.08 – Present **Research Scientist**
King Abdullah University of Science and Technology (KAUST), Kingdom of Saudi Arabia.

2022.08 – 2025.08 **Postdoctoral Fellow**
King Abdullah University of Science and Technology (KAUST), Kingdom of Saudi Arabia.

2022.01 – 2022.07 **Postdoctoral Fellow**
Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences, China.

Teaching

- **Guest lecturer** of Seismic Imaging course, KAUST, Saudi Arabia, 2024.
- **Guest lecturer** of Seismic Imaging course, KAUST, Saudi Arabia, 2023.

Supervision

- **Randy Harsuko** (co-supervisor), Ph.D. student in Geophysics, KAUST, since 2023. Machine learning for multi-tasks seismic processing and velocity model building.
- **Sixiu Liu** (co-supervisor), Ph.D. student in Geophysics, KAUST, since 2023. Machine learning for self-supervised denoising and full-waveform inversion.
- **Diogo Duarte** (co-supervisor), visiting Ph.D. student in Geophysics, KAUST, since 2024. Physics-informed neural network for wavefield representation.
- **Xiao Ma** (co-supervisor), Ph.D. student in Geophysics, KAUST, since 2024. Physics-informed neural operator for microseismic location.
- **Hao Zhang** (co-supervisor), Ph.D. student in Geophysics, KAUST, since 2025. The applications of generative diffusion models in inverse problem.

Services

- Associate Editor for Geophysical Prospecting
- Special Issue Convenor for Physics-based machine learning advances in geophysical prospecting
- SEG's Early Career Community Member
- Session Chair for EAGE Annual Conference & Exhibition
- Session Chair for International Meeting for Applied Geoscience and Energy

- Reviewers for Geophysics, Geophysical Journal International, Geophysical Prospecting, Journal of Geophysical Research: Machine Learning and Computation, IEEE Transactions on Geoscience and Remote Sensing, IEEE Geoscience and Remote Sensing Letters, Computers & Geosciences, Frontiers in Earth Science, Journal of Applied Geophysics, Earth Science Informatics, Journal of Geophysics and Engineering, Scientific Reports, Chinese Journal of Geophysics, Acta Geophysica, Physical Review E, International Meeting for Applied Geoscience and Energy, etc.

Funded Projects

- Member, DeepWave Consortium, KAUST, Saudi Arabia, 2023-now.
- Project manage, High-efficiency viscoelastic anisotropic wavefield simulation method, 2021-2022
- Project member, High-precision seismic migration method for seafloor four-component OBN data (NSFC), 2021-2022
- Project member, Imaging domain multi-component viscoelastic least squares reverse-time migration method (NSFC), 2021-2022

Awards

- Pacemaker to Merit Student of University of Chinese Academy of Sciences (2021)
- The academic scholarship of University of Chinese Academy of Sciences (2018-2021)
- The outstanding cadres of Jilin University (2016-2017)
- The outstanding students of Jilin University (2013-2014)

Publications

Zhijun Cheng, Xiang Wang, Guojun Mao, Weijian Mao, and **Shijun Cheng** (2026). "Meta-transfer learning for efficient initialization of neural network wavefield solutions". In: **Journal of Applied Geophysics**, p. 106116.

Huan Song, **Shijun Cheng**, Huanhuan Tang, Wei Ouyang, and Weijian Mao (2026). "Adaptive Self-Supervised Surface-Related Multiple Suppression". In: **arXiv preprint arXiv:2604.27069 (Under review in IEEE Transactions on Geoscience and Remote Sensing)**.

Huanhuan Tang, **Shijun Cheng**, Weijian Mao, Haoran Zhang, and Yingying Zhang (2026). "Incorporating wave physical priors into diffusion models: A novel approach to seismic resolution enhancement". In: **Surveys in Geophysics**.

Huanhuan Tang, **Shijun Cheng**, Wei Ouyang, and Weijian Mao (2026). "Attenuating Multi-source Noise in DAS-VSP Data: A Self-Supervised Learning Approach with Enhanced Noise Extraction". In: **Big Data and Earth System**, p. 100062.

Shijun Cheng and Tariq Alkhalifah (2026a). "Generative wave propagator". In: **arXiv preprint arXiv:2607.0 (Under review in Geophysical Journal International)**.

Shijun Cheng and Tariq Alkhalifah (2026b). "Propagating the prior from far to near offset: A self-supervised diffusion framework for progressively recovering near-offsets of towed-streamer data". In: **arXiv preprint arXiv:2602.01909 (Under review in Geophysics)**.

Shijun Cheng, Randy Harsuko, Mohammad Hasyim Taufik, and Tariq Alkhalifah (2026). "Depth-progressive 3D seismic velocity model building via 2D generative diffusion models". In: **arXiv preprint arXiv:2610.14812 (Under review in Geophysics)**.

Shijun Cheng, Randy Harsuko, and Tariq Alkhalifah (2026a). "Shallow-to-deep velocity model building via diffusion models-Part I: Method and Proof of concept". In: **arXiv preprint arXiv:2609.26454 (Under review in Geophysics)**.

Shijun Cheng, Randy Harsuko, and Tariq Alkhalifah (2026b). "Shallow-to-deep velocity model building via diffusion models-Part II: Realistic scenarios". In: **arXiv preprint arXiv:2609.26482 (Under review in Geophysics)**.

Shijun Cheng, Xinru Mu, and Tariq Alkhalifah (2026). "Physics-informed conditional diffusion model for generalizable elastic wave-mode separation". In: **Geophysics (Acepted)**.

Shijun Cheng, Mohammad Hasyim Taufik, and Tariq Alkhalifah (2026). "Seismic wavefield solutions via physics-guided generative neural operator". In: **Geophysics**.

Zefeng Wang, **Shijun Cheng**, Weijian Mao, Wei Ouyang, and Huanhuan Tang (2026). "Meta-learning-enhanced implicit full waveform inversion". In: **Geophysical Prospecting**.

Lingli Gao, Weijian Mao, Qianru Xu, Wei Ouyang, Shaokang Yang, and **Shijun Cheng*** (2025). "Inhomogeneous plane waves in attenuative anisotropic porous media". In: **Geophysical Journal International**.

Randy Harsuko, **Shijun Cheng**, and Tariq Alkhalifah (2025a). "Propagating the prior from shallow to deep with a pre-trained velocity-model Generative Transformer network". In: **Journal of Geophysical Research: Machine Learning and Computation**.

Randy Harsuko, **Shijun Cheng**, and Tariq Alkhalifah (2025b). "Velocity model building with uncertainty quantification using a multi-modal autoregressive generative network". In: **Under review in Geophysical Journal International**.

Xinru Mu, **Shijun Cheng**, and Tariq Alkhalifah (2025). "Separationpinn: Physics-informed neural networks for seismic p-and s-wave mode separation". In: **IEEE Transactions on Geoscience and Remote Sensing**.

Huanhuan Tang, **Shijun Cheng**, Wuqun Li, and Weijian Mao (2025). "An EDCC-EMD analysis-based network for DAS VSP data denoising in frequency domain". In: **Petroleum Science**.

Shijun Cheng and Tariq Alkhalifah (2025a). "DiffPINN: Generative diffusion-initialized physics-informed neural networks for accelerating seismic wavefield representation". In: **arXiv preprint arXiv:2506.00471 (Under review in Geophysics)**.

Shijun Cheng and Tariq Alkhalifah (2025b). "Meta learning for improved neural network wavefield solutions". In: **Surveys in Geophysics**.

Shijun Cheng and Tariq Alkhalifah (2025c). "Multi-frequency wavefield solutions for variable velocity models using meta-learning enhanced low-rank physics-informed neural network". In: **Geophysical Journal International**.

Shijun Cheng, Randy Harsuko, and Tariq Alkhalifah (2025). "A generative foundation model for an all-in-one seismic processing framework". In: **Surveys in Geophysics**.

Shijun Cheng, Ning Wang, and Tariq Alkhalifah (2025). "Self-supervised surface-related multiple suppression with multidimensional convolution". In: **arXiv preprint arXiv:2505.00419 (Under review in Geophysics)**.

Shijun Cheng, Haoran Zhang, and Tariq Alkhalifah (2025). "Self-supervised Seismic Resolution Enhancement". In: **IEEE Transactions on Geoscience and Remote Sensing**.

Qingchen Zhang, **Shijun Cheng***, Wei Chen, and Weijian Mao (2025). "Well-and Structure-Constrained Initial Velocity Building for Full-Waveform Inversion via a Generative Diffusion Model". In: **IEEE Transactions on Geoscience and Remote Sensing**.

Sixiu Liu, **Shijun Cheng**, and Tariq Alkhalifah (2024). "Gabor-based learnable sparse representation for self-supervised denoising". In: **IEEE Transactions on Geoscience and Remote Sensing**.

Xingchen Shi, **Shijun Cheng***, Weijian Mao, and Wei Ouyang (2024). "Generative Diffusion Model for Seismic Imaging Improvement of Sparsely Acquired Data and Uncertainty Quantification". In: **IEEE Transactions on Geoscience and Remote Sensing** 62, pp. 1–17.

Shijun Cheng and Tariq Alkhalifah (2024a). "Discovery of physically interpretable wave equations". In: **Surveys in Geophysics**.

Shijun Cheng and Tariq Alkhalifah (2024b). "Robust data driven discovery of a seismic wave equation". In: **Geophysical Journal International** 236.1, pp. 537–546.

Shijun Cheng, Zhiyao Cheng, Chao Jiang, Weijian Mao, and Qingchen Zhang (2024). "An effective self-supervised learning method for attenuating various types of seismic noise". In: **Geophysics** 89.6, pp. 1–78.

Shijun Cheng, Randy Harsuko, and Tariq Alkhalifah (2024). "Meta-processing: a robust framework for multi-tasks seismic processing". In: **Surveys in Geophysics**.

Shijun Cheng, Yi Wang, Qingchen Zhang, Randy Harsuko, and Tariq Alkhalifah (2024). "A self-supervised learning framework for seismic low-frequency extrapolation". In: **Journal of Geophysical Research: Machine Learning and Computation** 1.3, e2024JH000157.

Wei Ouyang, Quan Liang, Weijian Mao, and **Shijun Cheng*** (2023). "Asymptotic linearized inversion for acoustic variable-tilt transversely isotropic media through elliptic perturbation analysis". In: **Wave Motion** 120, p. 103161.

Wei Ouyang, Quan Liang, Weijian Mao, and **Shijun Cheng** (2023). "Multiparameter acoustic inversion for variable-tilt transversely isotropic media with generalized Radon transform". In: **IEEE Transactions on Geoscience and Remote Sensing**.

Huanhuan Tang, **Shijun Cheng**, Wuqun Li, and Weijian Mao (2023). "Simultaneous reconstruction and denoising for das-vsp seismic data by rru-net". In: **Frontiers in Earth Science** 10, p. 993465.

Huanhuan Tang, **Shijun Cheng**, Huan Song, and Weijian Mao (2023). "3D9C seismic data reconstruction with multi-scale convolution neural network". In: **Journal of Applied Geophysics** 216, p. 105163.

Shijun Cheng, Xingchen Shi, Weijian Mao, Tariq A Alkhalifah, Tao Yang, Yuzhu Liu, and Heping Sun (2023). "Elastic seismic imaging enhancement of sparse 4c ocean-bottom node data using deep learning". In: **IEEE Transactions on Geoscience and Remote Sensing** 61, pp. 1–14.

Lei Zhao, Weijian Mao, Zongbao Liu, and **Shijun Cheng** (2023). "Research on the differential tectonic-thermal evolution of Longmaxi shale in the southern Sichuan Basin". In: **Advances in Geo-Energy Research**.

Meng Du, **Shijun Cheng**, and Weijian Mao (2022). "Deep-learning-based seismic variable-size velocity model building". In: **IEEE Geoscience and Remote Sensing Letters** 19, pp. 1–5.

Qingqing Li, Qingchen Zhang, Qizhen Du, and **Shijun Cheng** (2022). "Well-guided multisource elastic full-waveform inversion". In: **IEEE Transactions on Geoscience and Remote Sensing** 60, pp. 1–14.

Quan Liang, Wei Ouyang, Weijian Mao, **Shijun Cheng**, and Xuelei Li (2022). "Born scattering integral, scattering radiation pattern, and generalized radon transform inversion in acoustic tilted transversely isotropic media". In: **IEEE Transactions on Geoscience and Remote Sensing** 60, pp. 1–16.

Wei Ouyang, Quan Liang, Weijian Mao, and **Shijun Cheng** (2021a). "Linearized scattering problem in acoustic transversely isotropic media with a vertical symmetry axis". In: **Wave Motion** 104, p. 102753.

Wei Ouyang, Quan Liang, Weijian Mao, and **Shijun Cheng** (2021b). "Multi-parameter true-amplitude generalized radon transform inversion for acoustic transversely isotropic media with a vertical symmetry axis". In: **IEEE Transactions on Geoscience and Remote Sensing** 60, pp. 1–14.

Shijun Cheng, WeiJian Mao, and Wei Ouyang (2021). "Plane wave propagation in viscoelastic attenuative VTI media based on fractional time derivatives". In: **Chinese Journal of Geophysics** 64.3, pp. 965–981.

Shijun Cheng, Weijian Mao, Qingchen Zhang, and Qianru Xu (2021). "Wave propagation in the poro-viscoelastic orthorhombic two-phase media: plane-wave theory and wavefield simulation". In: **Geophysical Journal International** 227.1, pp. 99–122.

Quan Liang, Wei Ouyang, Weijian Mao, and **Shijun Cheng** (2020). "Scattering pattern analysis and true-amplitude generalized Radon transform migration for acoustic transversely isotropic media with a vertical axis of symmetry". In: **Geophysical Prospecting** 68.4, pp. 1211–1227.

Conferences

Randy Harsuko, **Shijun Cheng**, and Tariq Alkhalifah (2026). "Neural Representation-Based Waveform Inversion: A Realistic Case Study". In: *87th EAGE Annual Conference & Exhibition*. Vol. 2026. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Huanhuan Tang, **Shijun Cheng**, Weijian Mao, Haoran Zhang, and Yingying Zhang (2026). "A Physics-Guided Self-Supervised Diffusion Model for Seismic Resolution Enhancement". In: *87th EAGE Annual Conference & Exhibition*. Vol. 2026. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Mohammad Hasyim Taufik, **Shijun Cheng**, and Tariq Alkhalifah (2026). "Latent Space Optimization in Neural Operator-Based Waveform Inversion". In: *87th EAGE Annual Conference & Exhibition*. Vol. 2026. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Shijun Cheng and Tariq Alkhalifah (2026a). "Learning to propagate: Seismic wavefield modeling with diffusion models". In: *Sixth International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Shijun Cheng and Tariq Alkhalifah (2026b). "Observation-conditioned diffusion models for surrogate seismic forward operator". In: *Sixth International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Shijun Cheng and Tariq Alkhalifah (2026c). "Unifying seismic forward and inverse operators with a joint guided diffusion model". In: *Sixth International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Shijun Cheng, Randy Harsuko, and Tariq Alkhalifah (2026). "Generative Diffusion Models for Shallow-to-Deep Seismic Velocity Model Building". In: *87th EAGE Annual Conference & Exhibition*. Vol. 2026. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Shijun Cheng, Randy Harsuko, Mohammad Hasyim Taufik, and Tariq Alkhalifah (2026). "Depth-progressive 3D seismic velocity model building via 2D generative diffusion models". In: *87th EAGE Annual Conference & Exhibition*. Vol. 2026. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Tariq Alkhalifah, Randy Harsuko, and **Shijun Cheng** (2025). "Using Generative Models for Seismic Processing and Velocity Model Building". In: *86th EAGE Annual Conference & Exhibition (Workshop Programme)*. Vol. 2025. 1. European Association of Geoscientists & Engineers, pp. 1–3.

Randy Harsuko, **Shijun Cheng**, and Tariq Alkhalifah (2025a). "Synthesizing realistic-scale seismic velocity models using a spatially-aware generative model". In: *86th EAGE Annual Conference & Exhibition (including the Workshop Programme)*. Vol. 2025. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Randy Harsuko, **Shijun Cheng**, and Tariq Alkhalifah (2025b). "Using a pretrained neural representation of seismic data for generative velocity model prediction". In: *86th EAGE Annual Conference & Exhibition (including the Workshop Programme)*. Vol. 2025. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Xiao Ma, **Shijun Cheng**, and Tariq Alkhalifah (2025). "Microseismic Source Inversion Based on Neural Operator with Physics Constrain". In: *86th EAGE Annual Conference & Exhibition*. Vol. 2025. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Shijun Cheng and Tariq Alkhalifah (2025). "Accelerating seismic wavefield representation via latent diffusion-initialized physics-informed neural networks". In: *Fifth International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Shijun Cheng, Randy Harsuko, and Tariq Alkhalifah (2025). "Multi-task seismic processing using generative diffusion models". In: *86th EAGE Annual Conference & Exhibition (including the Workshop Programme)*. Vol. 2025. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Shijun Cheng, Xinru Mu, and Tariq Alkhalifah (2025). "Toward generalizable elastic wave separation: A physics-informed conditional diffusion approach". In: *Fifth International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Shijun Cheng, Mohammad H. Taufik, and Tariq Alkhalifah (2025). "A generative neural operator for seismic wavefield representation". In: *86th EAGE Annual Conference & Exhibition (including the Workshop Programme)*. Vol. 2025. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Shijun Cheng, Ning Wang, and Tariq Alkhalifah (2025). "Self-supervised free-surface multiple suppression". In: *86th EAGE Annual Conference & Exhibition (including the Workshop Programme)*. Vol. 2025. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Randy Harsuko, **Shijun Cheng**, and Tariq Alkhalifah (2024a). "Multi-conditional seismic velocity synthesis using an autoregressive Transformer network". In: *Fourth International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Randy Harsuko, **Shijun Cheng**, and Tariq Alkhalifah (2024b). "VelocityGPT: A Sequential Depth-Wise Velocity Model Generator in an Autoregressive Framework". In: *85th EAGE Annual Conference &*

Exhibition (including the Workshop Programme). Vol. 2024. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Sixiu Liu, Abdullah Alali, **Shijun Cheng**, and Tariq Alkhalifah (2024). "Integrating U-Net into full-waveform inversion for salt body building: A challenging case". In: *Fourth International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Sixiu Liu, **Shijun Cheng**, and Tariq Alkhalifah (2024). "Gabor-Based Learnable Sparse Representation for Self-Supervised Ground Roll Removal". In: *85th EAGE Annual Conference & Exhibition (including the Workshop Programme)*. Vol. 2024. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Shijun Cheng and Tariq Alkhalifah (2024). "Unit-constrained data-driven discovery of a wave equation". In: *Fourth International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Shijun Cheng, Randy Harsuko, and Tariq Alkhalifah (2024). "A Self-Supervised Learning Framework for Multi-Tasks Seismic Processing Via Meta-Learning". In: *85th EAGE Annual Conference & Exhibition (including the Workshop Programme)*. Vol. 2024. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Shijun Cheng, Haoran Zhang, and Tariq Alkhalifah (2024). "Seismic resolution enhancement with self-supervised learning". In: *Fourth International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Shijun Cheng and Tariq Alkhalifah (2023a). "D-SWE: Data-driven discovery of a seismic wave equation". In: *84th EAGE Annual Conference & Exhibition*. Vol. 2023. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Shijun Cheng and Tariq Alkhalifah (2023b). "Meta-PINN: Meta learning for improved neural network wavefield solutions". In: *Third International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Shijun Cheng and Tariq Alkhalifah (2023c). "Meta-processing: Meta-learning based initialization for enhancement multi-seismic processing with limited training data". In: *Third International Meeting for Applied Geoscience & Energy*. Society of Exploration Geophysicists and American Association of Petroleum.

Shijun Cheng, Xingchen Shi, Weijian Mao, and Tariq Alkhalifah (2023). "Seismic imaging enhancement of sparse ocean-bottom node data using deep learning". In: *84th eage annual conference & exhibition*. Vol. 2023. 1. European Association of Geoscientists & Engineers, pp. 1–5.

Skills

- Advanced level in programming (FORTRAN, Python, MATLAB, and C).
- Skilled with deep-learning framework PyTorch.
- Skilled with high-performance computing in OpenMP, MPI, and CUDA.